DO NOW — RECALL

Lesson 147 · Firewalls, Proxies and Encryption: Layered

Name: _____ Class: _____ Date: _____

1. What attack floods a service with requests to reduce availability?

2. What is a foreign key used for?

3. Which storage type is volatile?

4. Which layer in TCP/IP handles IP routing?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A denial-of-service attack, potentially distributed (DDoS).

2. Model answer

Referencing a candidate key in another relation.

3. Model answer

RAM.

4. Model answer

The internet layer.

By the end, you can…

• Describe how a firewall enforces traffic rules using addresses, ports or connection state.

• Distinguish a forward proxy from a firewall and explain appropriate filtering or caching uses.

• Choose encryption and additional controls for confidentiality without assuming they stop every attack.

Vocabulary — know these terms

firewall — Rule-based network traffic access control.

proxy server — Server that acts as intermediary for client requests.

TLS — Protocol protecting transport of application data.

defence in depth — Overlapping safeguards against different failures.

NEW LEARNING

Read, annotate and explain

Filtering boundary

Firewall applies rules to traffic entering or leaving a network.

Mediating requests

Forward proxy can request resources on behalf of clients, cache and log access.

Protecting content

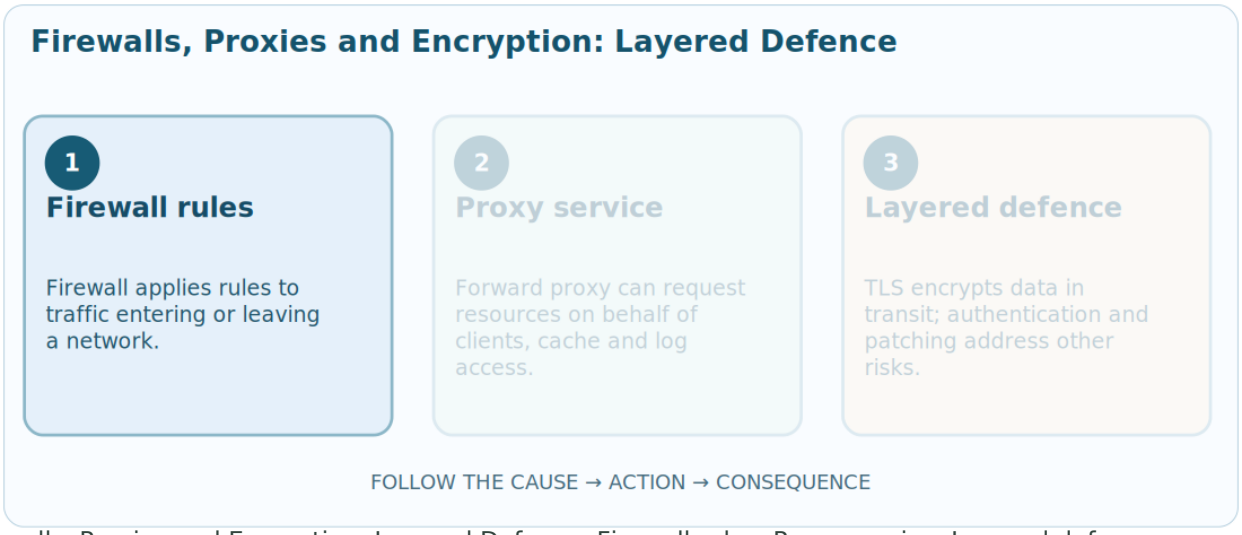
TLS encrypts data in transit; authentication and patching address other risks.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

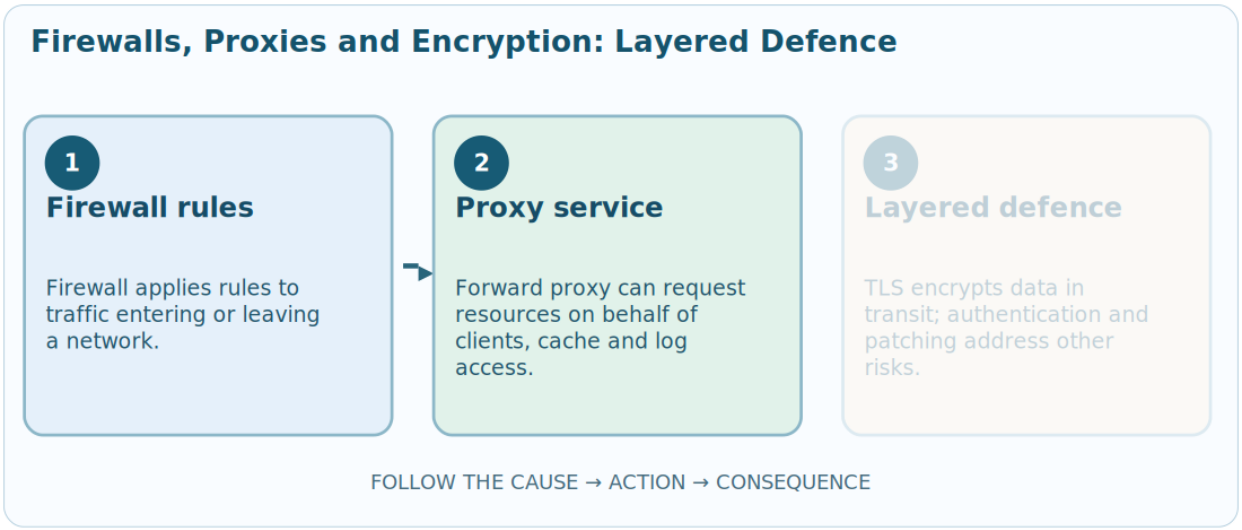
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Firewalls, Proxies and Encryption: Layered Defence: Firewall rules, Proxy service, Layered defence

New Learning

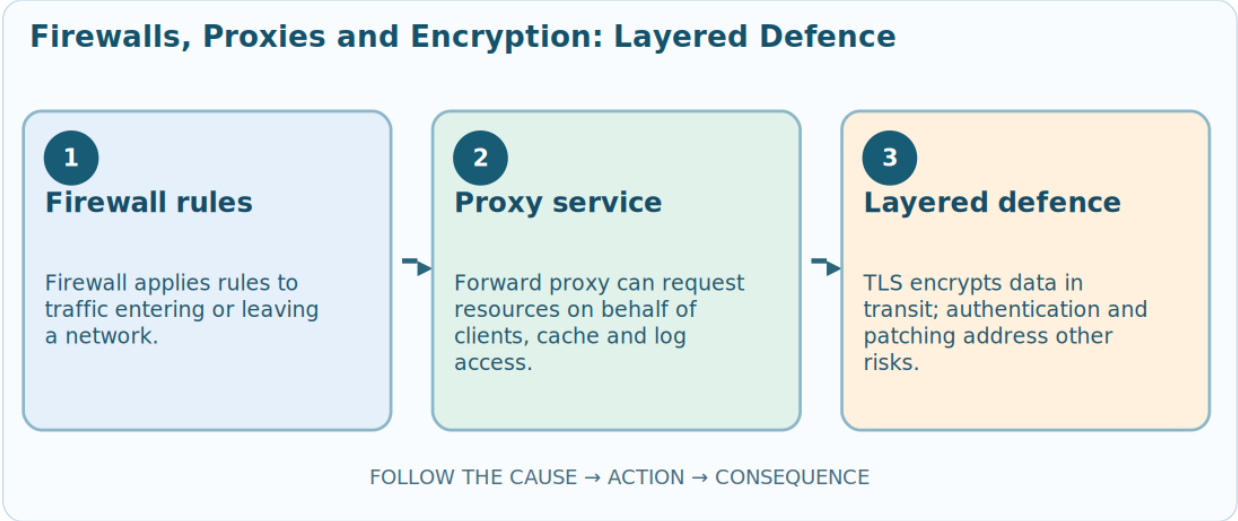


Firewalls, Proxies and Encryption: Layered Defence: Firewall rules, Proxy service, Layered defence

New Learning · original lesson diagrams

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New Learning



Firewalls, Proxies and Encryption: Layered Defence: Firewall rules, Proxy service, Layered defence

I DO — WORKED EXAMPLE

Follow the worked method

A school wants to limit outgoing web browsing while protecting pupils' login details over public Wi-Fi. Select appropriate controls.

Client web request

Identify the facts and make an initial prediction.

Proxy and firewall controls

Explain the mechanism or trace the intermediate stage.

HTTPS encrypted transfer

A managed web proxy can filter/log permitted requests; a firewall can restrict traffic flows; HTTPS/TLS encrypts the connection so local eavesdroppers cannot read credentials. Policy, MFA and endpoint protection address further threats.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school wants to limit outgoing web browsing while protecting pupils' login details over public Wi-Fi. Select

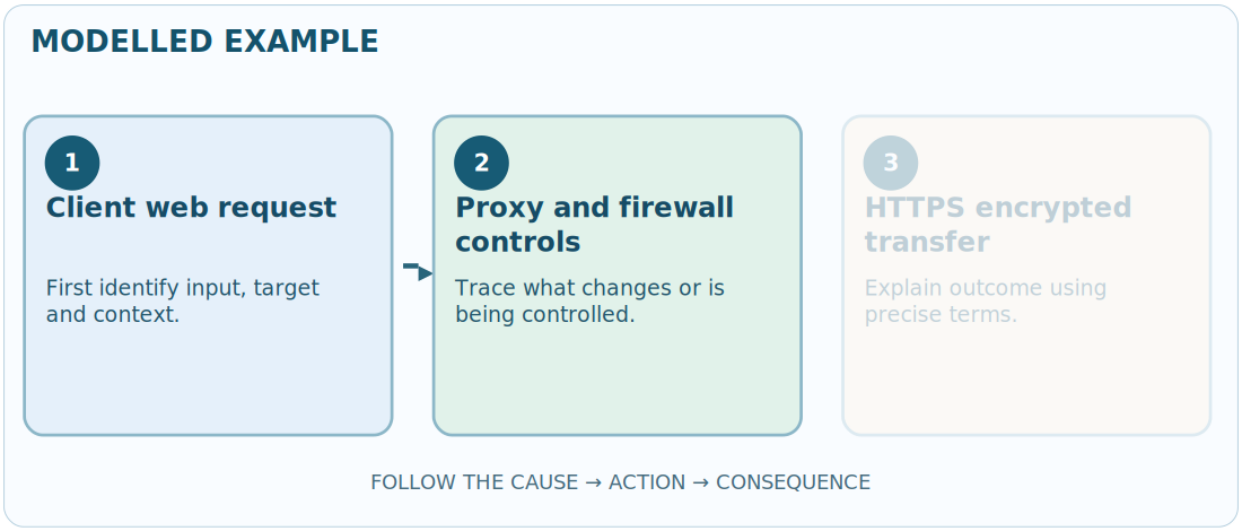
Identify the facts and make an initial prediction.



MODELLED EXAMPLE: Client web request, Proxy and firewall controls, HTTPS encrypted transfer

A school wants to limit outgoing web browsing while protecting pupils' login details over public Wi-Fi. Select

Identify the facts and make an initial prediction.



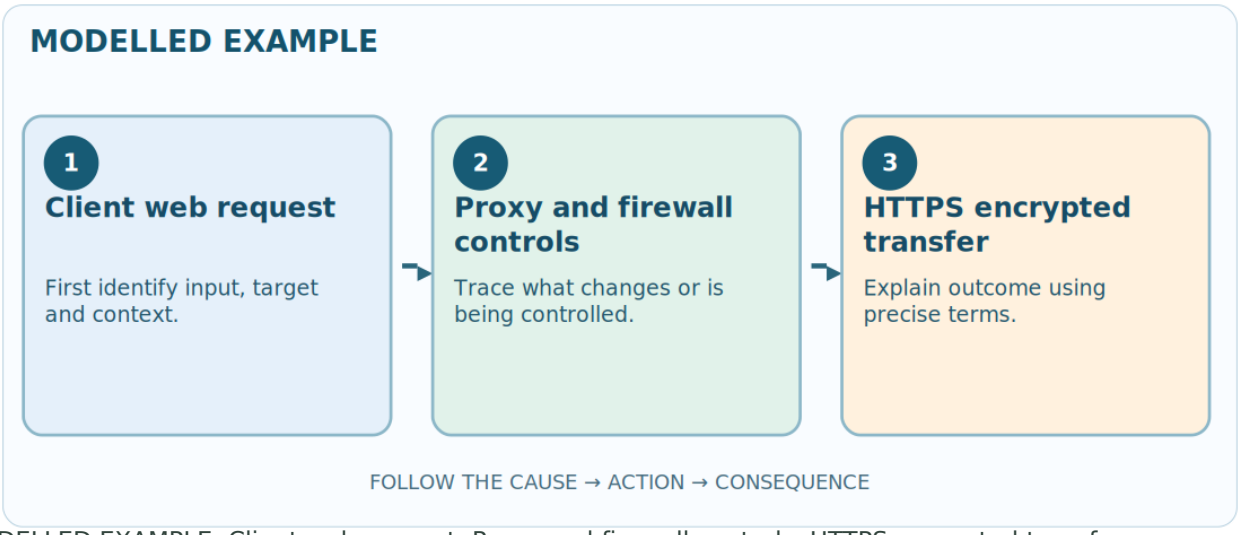
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I Do · original lesson diagrams

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MODELLED EXAMPLE: Client web request, Proxy and firewall controls, HTTPS encrypted transfer

WE DO — GUIDED PRACTICE

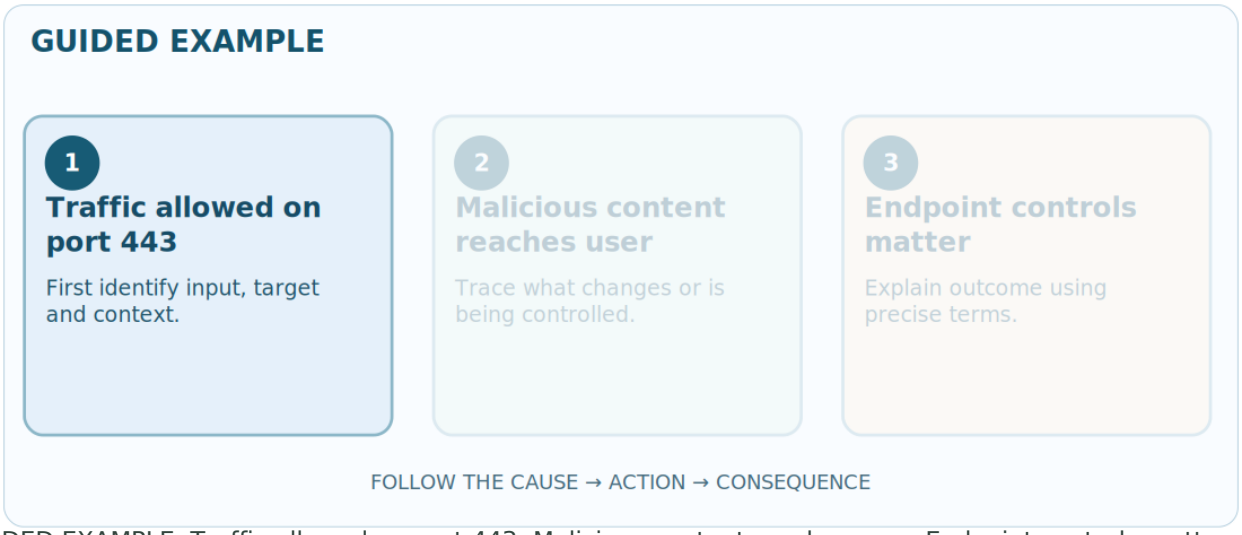
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A small business has a firewall allowing HTTPS but an employee clicks a malicious attachment. Why was the fire

Identify the facts and make an initial prediction.



GUIDED EXAMPLE: Traffic allowed on port 443, Malicious content reaches user, Endpoint controls matter

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Traffic allowed on port 443 Identify the facts and make an initial prediction.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State functions of firewall, forward proxy and TLS, one network example each. [6]

Question 2

Explain why encrypted traffic is not the same as authenticated trustworthy content. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Recommend a layered defence for a sixth-form network with remote learners, justifying four different controls. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A forward proxy and a firewall necessarily perform identical tasks.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

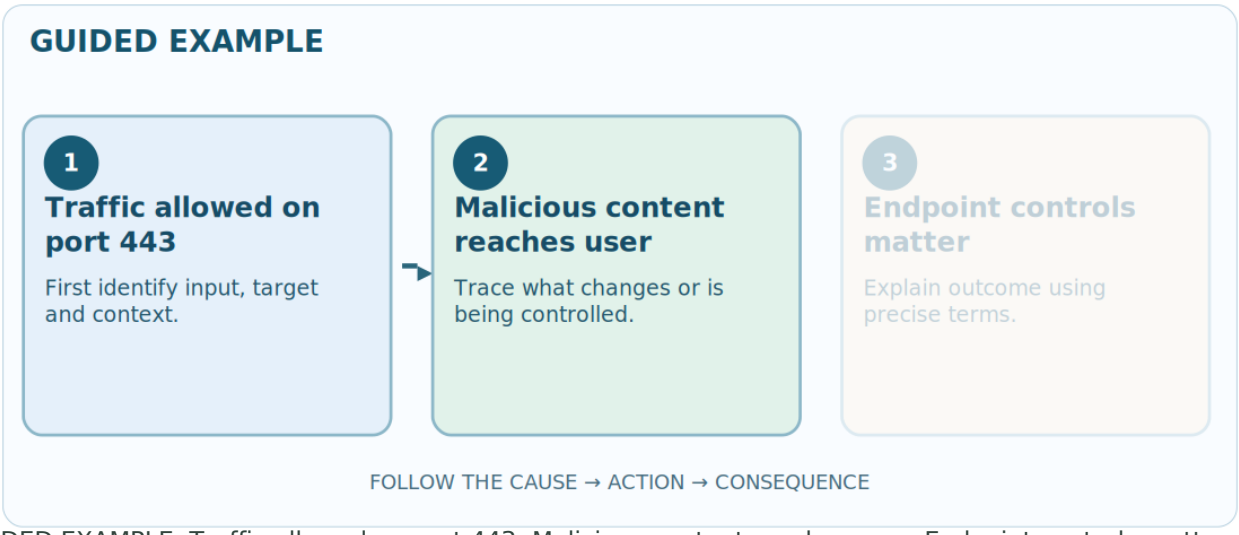
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

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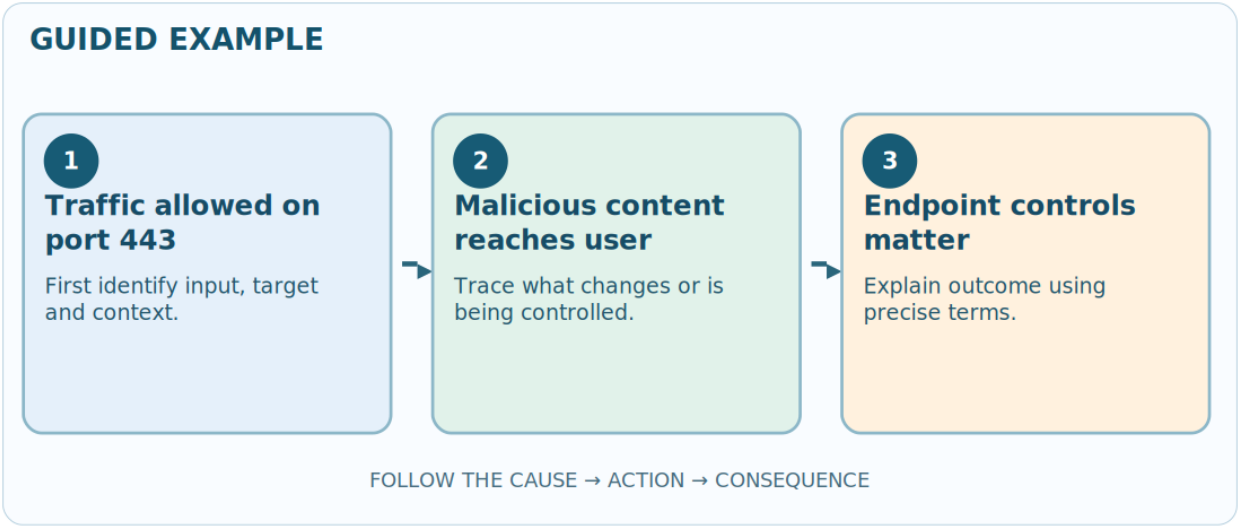
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GUIDED EXAMPLE: Traffic allowed on port 443, Malicious content reaches user, Endpoint controls matter

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WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Malicious content reaches user

Explain the mechanism or trace the intermediate stage.

Endpoint controls matter

An allowed connection or email can carry malicious content. Endpoint malware detection, attachment scanning, patching, permissions and user training complement network filtering. A proxy may help inspect or block risky destinations.

IMPROVE: Explain one correction to your own reasoning.